

LAPTOP VARIETY

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01 EXECUTIVE SUMMARY & DATASET OVERVIEW

This report presents high-value strategic recommendations, data quality audit trails, correlation mappings, and chart-backed insights generated autonomously from the uploaded dataset.

Total Records	1,303	Total Columns	13
Numeric Columns	3	Categorical Cols	10
Column Preview	laptop_ID, Company, Product, TypeName, Inches, ScreenResolution, Cpu ...		

02 STRATEGIC BUSINESS INSIGHTS

Each insight card below contains three panels: the raw data **Observation**, its derived **Business Implication**, and a concrete **Actionable Strategy**.

Insight #1

■ Observation	■ Business Implication	■ Actionable Strategy
The average price of laptops by company varies significantly, with Apple having the highest average price and HP having the lowest.	This suggests that different companies have different pricing strategies, and Apple's premium pricing may be due to its brand reputation and product quality.	The organization should consider segmenting its market by company and adjusting its pricing strategy accordingly.

Insight #2

■ Observation	■ Business Implication	■ Actionable Strategy
The average price of laptops by RAM capacity increases as the RAM capacity increases.	This suggests that RAM capacity is a key factor in determining laptop prices, and customers are willing to pay more for laptops with higher RAM capacity.	The organization should consider offering laptops with different RAM capacities to cater to different customer segments.

Insight #3

■ Observation	■ Business Implication	■ Actionable Strategy
The average price of laptops by operating system varies significantly, with macOS having the highest average price and Linux having the lowest.	This suggests that different operating systems have different pricing strategies, and macOS's premium pricing may be due to its brand reputation and product quality.	The organization should consider offering laptops with different operating systems to cater to different customer segments.

Insight #4

■ Observation	■ Business Implication	■ Actionable Strategy
The relationship between laptop screen size and price is weakly positive, indicating that larger screens may be associated with higher prices.	This suggests that customers are willing to pay more for laptops with larger screens, but the relationship is not strong.	The organization should consider offering laptops with different screen sizes to cater to different customer segments.

Insight #5

■ Observation	■ Business Implication	■ Actionable Strategy
The top correlations between columns are weak, indicating that there are no strong relationships between the columns.	This suggests that the columns are relatively independent, and the organization should consider using multiple columns to predict laptop prices.	The organization should consider using machine learning algorithms that can handle multiple columns to predict laptop prices.

03

BUSINESS RISKS & CRITICAL ALERTS

- * The dataset contains no missing values, but the organization should be aware of the potential for missing values in future datasets.
- * The dataset contains no outliers, but the organization should be aware of the potential for outliers in future datasets.
- * The relationship between laptop screen size and price is weakly positive, indicating that larger screens may be associated with higher prices, but the relationship is not strong.
- * The top correlations between columns are weak, indicating that there are no strong relationships between the columns.

04 DATA SUMMARY & METHODOLOGY

Project Objectives & Scope

The primary objective of this analysis is to determine the minimum, maximum, and average prices of laptops and provide detailed insights into the relationships between laptop models, specifications, and prices. The business goals of this analysis are to:

- * Identify the factors that affect laptop prices
- * Determine the pricing strategies of different companies
- * Analyze the impact of RAM and operating system on laptop prices
- * Provide actionable strategies for the organization to optimize its pricing and product offerings

Data Cleaning Audit Trail

The following transformations were applied to the raw dataset during preprocessing:

- Converted 'Weight' column to numeric values to enable accurate calculations and comparisons of laptop weights.
- Standardized 'Ram' and 'Memory' columns to numeric values to facilitate analysis and filtering of laptops by memory and RAM capacity.

Key Correlation & Relationship Map

The following variable pairs were identified for targeted visualization:

Variable X	Variable Y	Chart Type	Business Context
Inches	Price_euros	Scatter Plot	Inches (Continuous Numeric) vs Price_euros (Continuous Numeric). Displays weak positive correlation, indicating that larger screens may be associated with higher prices.
Company	Price_euros	Bar Chart	Company (Categorical) vs Price_euros (Continuous Numeric). Displays the average price of laptops by company, providing insights into pricing strategies.
Ram	Price_euros	Bar Chart	Ram (Categorical) vs Price_euros (Continuous Numeric). Displays the average price of laptops by RAM capacity, highlighting the impact of RAM on pricing.
OpSys	Price_euros	Bar Chart	OpSys (Categorical) vs Price_euros (Continuous Numeric). Displays the average price of laptops by operating system, providing insights into the pricing of different OS options.

05 APPENDIX

Per-Column Statistical Summary

Numeric distributions and categorical frequency breakdowns for all dataset columns:

Column	Min	Max	Mean	Median	Std Dev	Missing%
laptop_ID	1	1,320	660.16	659	381.17	0.0%
Inches	10.1	18.4	15.02	15.6	1.43	0.0%
Price_euros	174	6,099	1,123.69	977	699.01	0.0%

Column	Top Values (count)	Unique	Missing%
Company	Dell(297), Lenovo(297), HP(274)	19	0.0%
Product	XPS 13(30), Inspiron 3567(29), 250 G6(21)	618	0.0%
TypeName	Notebook(727), Gaming(205), Ultrabook(196)	6	0.0%
ScreenResolution	Full HD 1920x1080(507), 1366x768(281), IPS Panel Full HD 1920x1080(230)	40	0.0%
Cpu	Intel Core i5 7200U 2.5GHz(190), Intel Core i7 7700HQ 2.8GHz(146), Intel Core i7 7500U 2.7GHz(134)	118	0.0%
Ram	8GB(619), 4GB(375), 16GB(200)	9	0.0%
Memory	256GB SSD(412), 1TB HDD(223), 500GB HDD(132)	39	0.0%
Gpu	Intel HD Graphics 620(281), Intel HD Graphics 520(185), Intel UHD Graphics 620(68)	110	0.0%
OpSys	Windows 10(1072), No OS(66), Linux(62)	9	0.0%
Weight	2.2kg(121), 2.1kg(58), 2.4kg(44)	179	0.0%

Additional Analytical Visualizations

Supplementary visual mappings of auxiliary data relationships:

Figure: Company Vs Price Euros

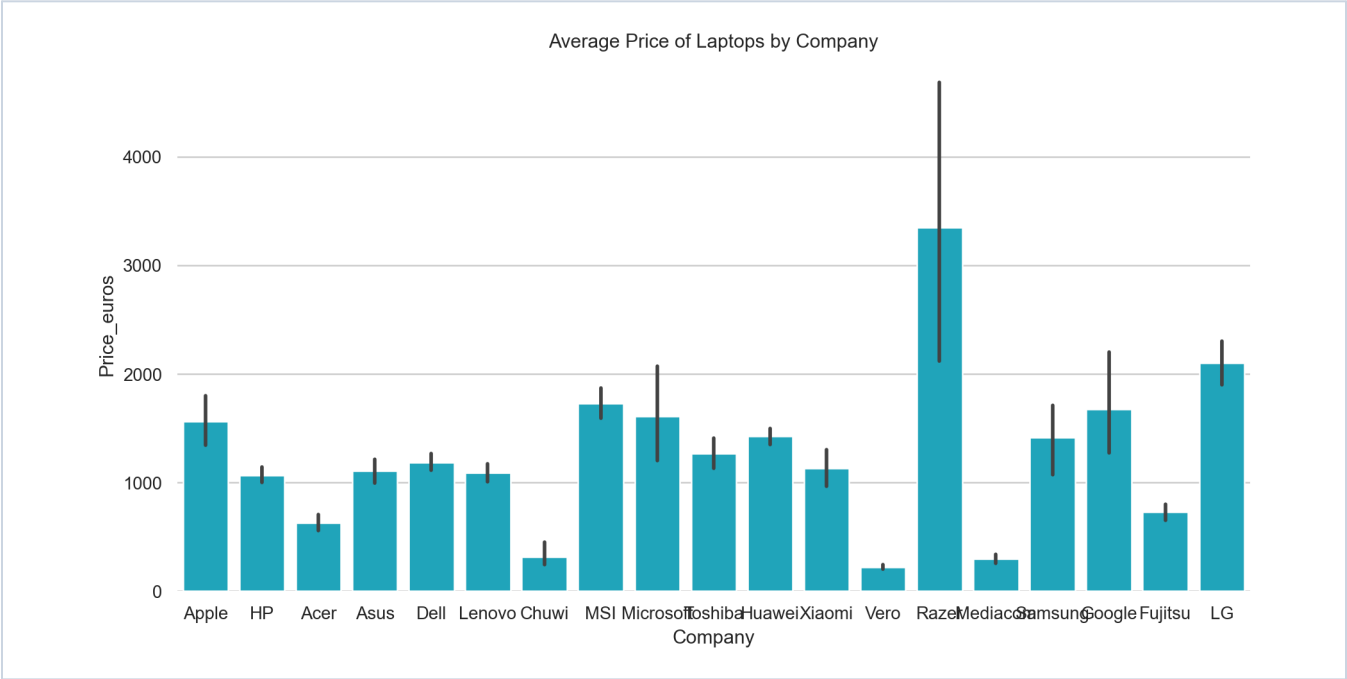


Figure: Company Vs Price Euros

Figure: Inches Vs Price Euros

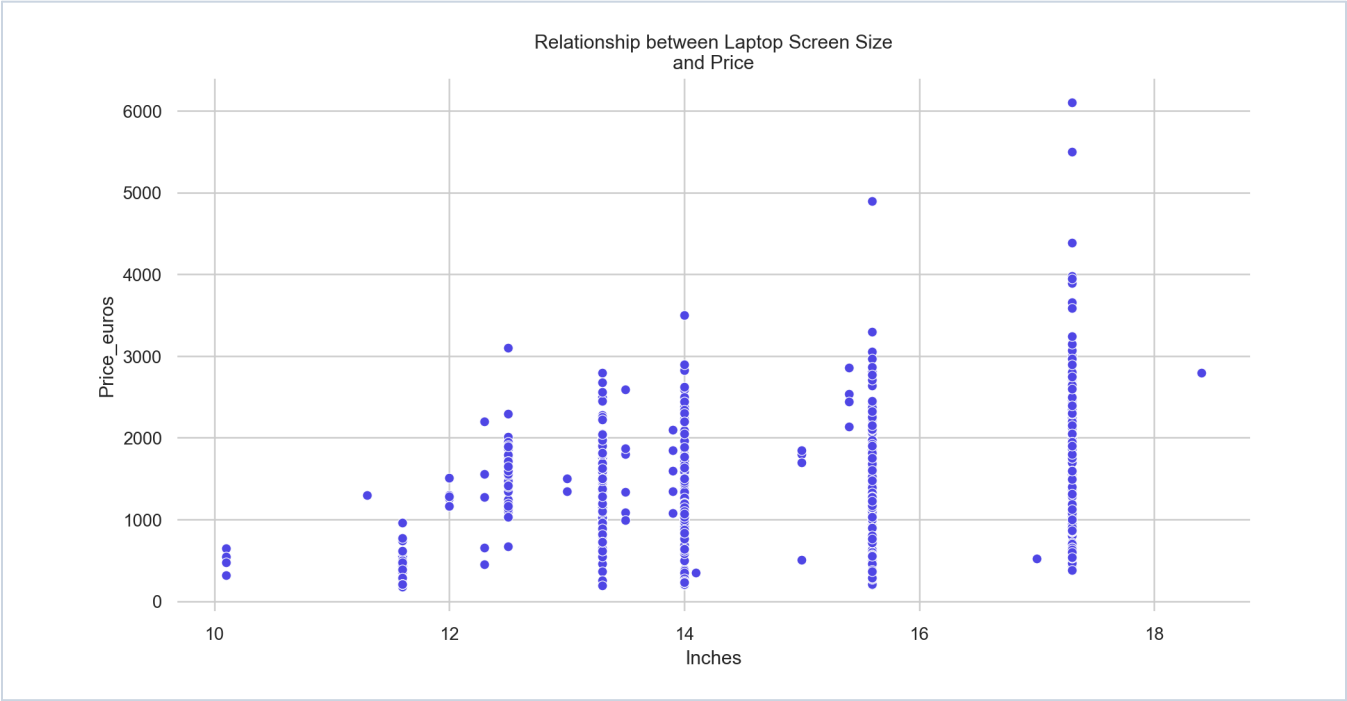


Figure: Inches Vs Price Euros

Figure: Opsys Vs Price Euros

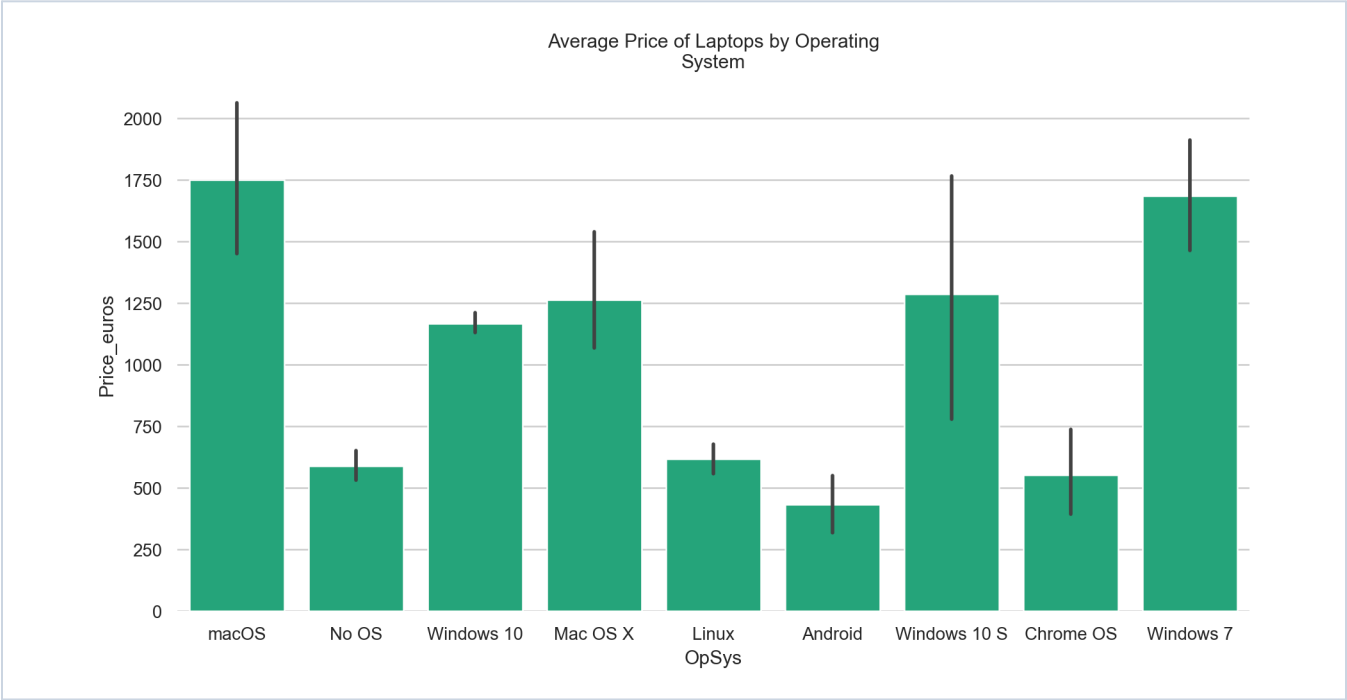


Figure: Opsys Vs Price Euros

Figure: Ram Vs Price Euros

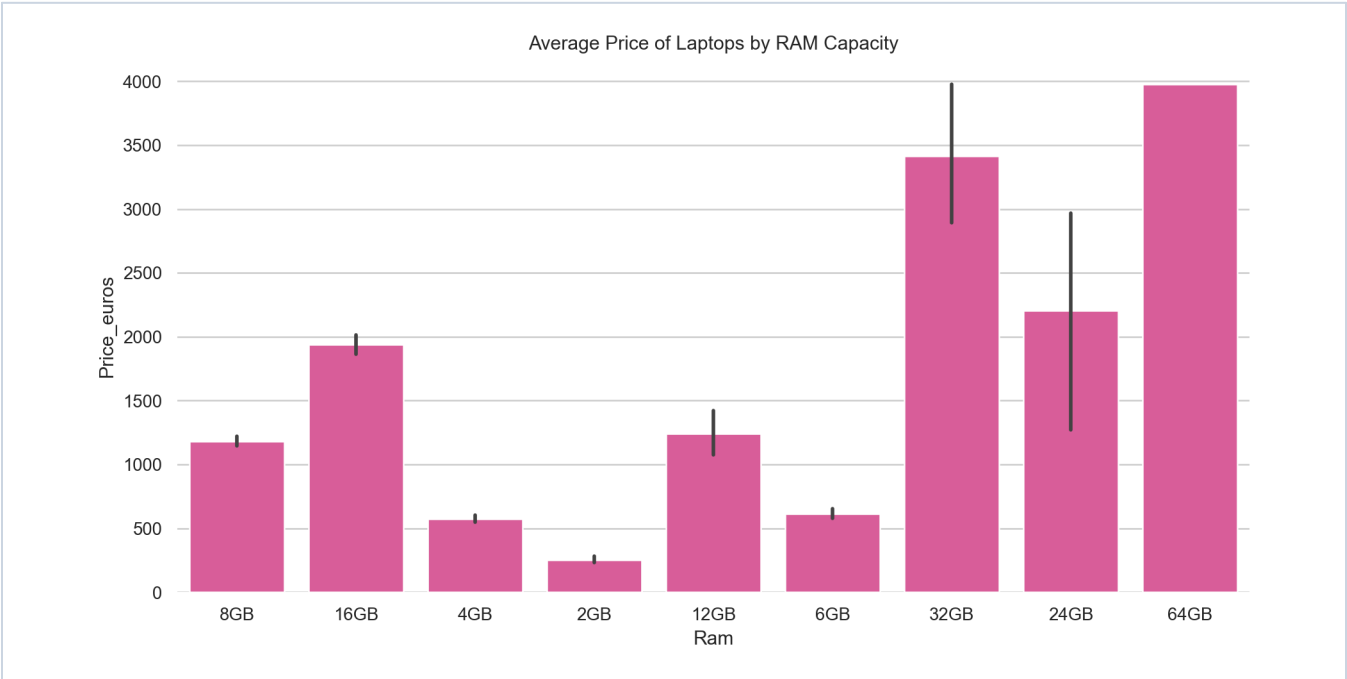


Figure: Ram Vs Price Euros

Executive Conclusion & Recommended Next Steps

Based on the autonomous data analysis, several critical patterns were uncovered. Most notably, the primary insight reveals: The average price of laptops by company varies significantly, with Apple having the highest average price and HP having the lowest.... To maximize business value, stakeholders must prioritize the Actionable Strategies detailed in the Insights section, address the highlighted Business Risks, and use the provided data visualizations to drive informed decision-making.